LayoverNow Risk Assessment (Q6 to Q8)

Top risks for the LayoverNow Chrome extension, with likelihood and impact scores and the software and infrastructure controls that would address them. As a client-side extension it has little dedicated hardware, so most controls are software or process based.

Scoring scale (1 to 5). Likelihood: 1 Rare, 2 Unlikely, 3 Possible, 4 Likely, 5 Almost certain. Impact: 1 Negligible, 2 Minor, 3 Moderate, 4 Major, 5 Severe (project failure or seriously misleading output). Risk Score = Likelihood x Impact (max 25).

Q6. Top risks

1. **Dependency on third-party AI and data APIs (V2).** V2's AeroAI Advisor and live enrichment depend on external AI and data APIs that can rate-limit, slow down, error out, or go offline, and whose keys carry cost.
2. **Manifest V3 / Chrome extension API changes.** Chrome periodically changes or deprecates extension APIs under Manifest V3, and a change can break loading or core functionality.
3. **Unreliable or hallucinated AI recommendations.** The AI advisor can return confidently wrong or fabricated assessments because AI outputs have no single checkable correct answer (the oracle problem).
4. **Inaccurate offline data or scoring approximation.** The local airport database can go stale, stopover-program data can change, and Haversine great-circle distance underestimates real routed travel time.
5. **Extension security and permission or data exposure.** As a browser extension, LayoverNow is exposed to permission over-reach and data-leak risks common to the extension ecosystem (cf. Obimbo et al., 2018).

Q7. Risk scores (Likelihood x Impact)

#	Risk	Likelihood	Impact	Risk Score
1	Dependency on third-party AI and data APIs (V2)	4	4	**16**
2	Manifest V3 / Chrome extension API changes	3	4	**12**
3	Unreliable or hallucinated AI recommendations	4	4	**16**
4	Inaccurate offline data or scoring approximation	3	3	**9**
5	Extension security and permission or data exposure	3	4	**12**

Ranked highest to lowest: Dependency on third-party AI and data APIs (V2) (16); Unreliable or hallucinated AI recommendations (16); Manifest V3 / Chrome extension API changes (12); Extension security and permission or data exposure (12); Inaccurate offline data or scoring approximation (9).

Q8. Software / hardware solutions

Dependency on third-party AI and data APIs (V2)

Software: response caching; automatic fallback to V1's deterministic ranking when an API is unavailable; request timeouts with retry and backoff; clear degraded-mode messaging;

quota and key-usage monitoring. Infrastructure: runs client-side, so no servers are required; keep the API key off the client via a thin proxy if cost or abuse is a concern.

Manifest V3 / Chrome extension API changes

Software: target stable documented MV3 APIs; avoid deprecated calls; feature-detect and degrade gracefully; track Chrome release notes; run a regression suite against each Chrome update.

Unreliable or hallucinated AI recommendations

Software: constrain prompts and validate AI output against deterministic checks (visa, season, layover length); never let the AI override hard constraints; show the V1 deterministic score alongside the AI verdict; apply robustness and metamorphic-style testing to the advisor (cf. Zhang et al., 2022).

Inaccurate offline data or scoring approximation

Software: version and date-stamp the local database with a scheduled refresh; unit-test the scoring and Haversine engine; label distances as estimates; validate stopover-program data against a source on each update.

Extension security and permission or data exposure

Software: follow least privilege and request only needed permissions; keep V1 search fully offline; use HTTPS only for V2 calls; store no sensitive personal data; apply a Content Security Policy; sanitize all inputs; review the permission set before each release.

References

Obimbo, C., Zhou, Y., & Nguyen, R. (2018). Analysis of vulnerabilities of web browser extensions. In 2018 International Conference on Computational Science and Computational Intelligence (CSCI) (pp. 116-119). IEEE. https://doi.org/10.1109/CSCI46756.2018.00029

Zhang, J. M., Harman, M., Ma, L., & Liu, Y. (2022). Machine learning testing: Survey, landscapes and horizons. IEEE Transactions on Software Engineering, 48(1), 1-36. https://doi.org/10.1109/TSE.2019.2962027